

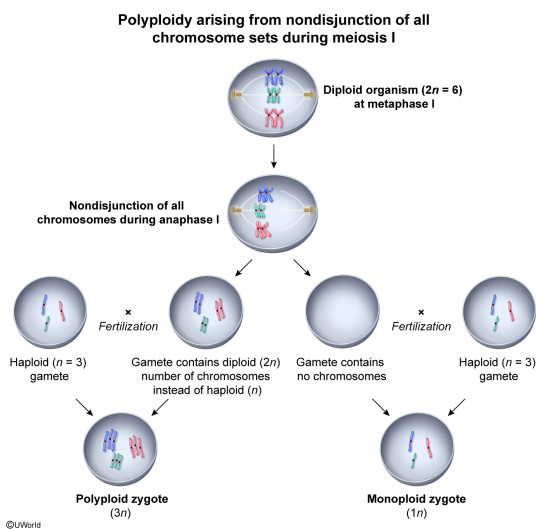
Formation of a polyploid organism would most likely occur through which mechanism?

- ☐ A. Duplication of chromosomes during S phase between meiosis I and II [19%]
- ☐ B. Unequal crossing over between misaligned homologous chromosomes [7%]
- ☐ C. Recombination between nonhomologous chromosomes [4%]
- ✓ ☒ D. Nondisjunction during anaphase I or II of meiosis [68%]

Correct

68% answered correctly

Explanation:



When the number of **homologous chromosomes** in the nucleus changes by an entire set, it is called a change in **ploidy**. Most organisms that reproduce sexually have **haploid** ($1n$) **gametes**, and thus the somatic cells for those organisms are **diploid** ($2n$).

In diploid organisms, **meiosis I** is considered a reductional division because it reduces the number of chromosome sets (ploidy) in the gamete by half. This prepares the gamete for fertilization, at which time the diploid number is restored. During **meiosis II**, an equal division (similar to **mitosis**) takes place, which results in the segregation of **sister chromatids** to each haploid gamete.

Some plant and animal species are **polyploid**, meaning that they have more than two *complete* sets of homologous chromosomes. Polyploid species can exist as triploid ($3n$), tetraploid ($4n$), or higher. Polyploid species are often sterile if they contain an odd number of chromosome sets due to an **inability** to form homologous pairs during meiosis.

This question asks for the most likely mechanism by which polyploid organisms arise. One mechanism is through **nondisjunction of all homologous chromosomes** in one or both parent organisms **during anaphase I or anaphase II of meiosis**. Polyploid organisms can also arise through duplication of the diploid genome *after* zygote formation due to failed **cytokinesis**.

(Choice A) Between meiosis I and II, a **brief interphase** occurs. DNA replication is inhibited during this period, so chromosomes would *not* be duplicated again during this brief interphase.

(Choices B and C) Unequal crossing over between misaligned homologous chromosomes and recombination between nonhomologous chromosomes are mechanisms that could result in the duplication or deletion of some portions of individual chromosomes, and would result in aneuploidy (missing or extra chromosomes or parts of chromosomes). This would *not* result in a complete duplication of all chromosomes, as in polyploidy.

Educational objective:

In organisms that reproduce sexually, a reductional division takes place during meiosis I in which the number of chromosomes sets is reduced by half. When errors such as nondisjunction occur during anaphase I or II, conditions such as polyploidy (additional sets of chromosomes) and aneuploidy (missing or extra chromosomes) can occur.

Time spent:0 sec

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Biology

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Genetics and Evolution